# Reversed-Process Complexity Circuit: Optoelectronic Locality Construction, Inverted Dual Space Manifolds, and T-Symmetry Violation

George Wagenknecht
Research Division on Optoelectronic Computation & Temporal Topologies

November 10, 2026

**Abstract**

We present a rigorous theoretical and physical formulation of the Reversed-Process (RP) Complexity Circuit, establishing a bridge between optoelectronic feedback systems, non-local causal boundaries, inverted dual-space manifolds, and explicit Time-Reversal ($T$) symmetry violation. By coupling a forward computational zone $F$ with a temporal mirror transformation $\mathcal{M}$ and a reversed-process oracle $\mathcal{O}$, the system constructs a closed recursive operator network $X_{n+1} = \mathcal{O}(\mathcal{M}(F(X_n)))$. We derive the differential equations governing the optoelectronic driver dynamics, establish topological locality constraints under relativistic and sub-light signaling bounds, formulate the dual-space manifold inversion via symplectic mapping, and prove that non-conservative feedback loops in the temporal mirror induce macroscopic $T$-symmetry violation.

## 1 Introduction and Architectural Foundations

Conventional computational architectures execute operations along a strictly forward-directed temporal timeline, adhering to thermodynamic arrow constraints and local causality principles. The Reversed-Process (RP) Complexity Circuit modifies this paradigm by introducing a dual-zone temporal mirror topology wherein terminal computational states are mapped onto a reversed processing sequence via an optoelectronic viewport inversion process [1].

Let $\mathcal{X}$ denote the input state space, $\mathcal{S}_f$ the forward state space, $\mathcal{S}_\tau$ the mirrored temporal state space, and $\mathcal{S}_o$ the oracle output domain. The circuit operates through the composition of mappings:

$$F : \mathcal{X} \to \mathcal{S}_f, \quad \mathcal{M} : \mathcal{S}_f \to \mathcal{S}_r, \quad R : \mathcal{S}_r \to \mathcal{S}_o, \quad \mathcal{O} : \mathcal{S}_o \times \mathcal{H}_n \to \mathcal{O}_{out}, \quad \mathcal{V} : \mathcal{O}_{out} \to \mathcal{X} \tag{1}$$

where $\mathcal{V}$ represents the optoelectronic viewport inversion operator, mapping the reversed-process oracle output back into the forward domain as the initial condition for the subsequent cycle.

The complete system evolution follows the nonlinear recurrence relation:

$$X_{n+1} = \mathcal{V}(\mathcal{O}(\mathcal{R}(\mathcal{M}(F(\mathcal{D}(I)))))) \tag{2}$$

where $\mathcal{D}$ denotes optical acquisition and $I$ represents the optical intensity distribution. This closed-loop configuration forces the terminal state of cycle $n$ to serve as the initiating parameter of cycle $n + 1$, generating non-trivial recursive state trajectories that deviate from standard feed-forward execution.

## 2 Optoelectronic Implementation and Driver Dynamics

The physical realization of the RP circuit relies on optoelectronic hardware comprising an infrared laser source, a spatial beam splitter/reflector, dual InGaAs photodetector arrays (Detectors A and B), parallel drivers, and an analog IF/THEN integration link [2].

### 2.1 Driver A and B Differential Mechanics

Driver A operates as a randomized witness test driven by optical shot noise and photon fluctuation statistics. The intensity difference between Detector A and Detector B yields an instantaneous photocurrent $i_{\text{det}}(t)$. The randomized witness variable $r_n$ is derived by hashing the spatial motion energy mask $E(x, y, t)$:

$$r_n = 2 + (\text{SHA-256}(E) \pmod{x_n - 3}) \tag{3}$$

The witness evaluation function yields $A = 1$ if $r_n$ is a Miller-Rabin witness for composite instance $x_n$, and $A = 0$ otherwise.

Conversely, Driver B implements a deterministic check $c(x_n)$ charging an integrating capacitor $C$ through a resistance $R$. The voltage across the capacitor $V_C(t)$ obeys the first-order differential equation:

$$\frac{dV_C}{dt} + \frac{1}{RC} V_C(t) = \frac{V_s}{RC} c(x_n) \tag{4}$$

whose exact solution for a frame period $T_f$ is:

$$V_C(t) = V_s \left(1 - e^{-\frac{t}{RC}}\right) c(x_n) \tag{5}$$

An NPN transistor connected as an emitter follower provides an output voltage $V_B(t) = \max(0, V_C(t) - V_{BE})$, where $V_{BE} = 0.7$ V. For the logic HIGH threshold $V_{IH} = 3.5$ V with a supply $V_s = 5$ V, the required logic transition time is given by:

$$t_{\text{logic}} = RC \ln\left(\frac{V_s}{V_s - V_{IH} - V_{BE}}\right) \approx 1.83\, RC \tag{6}$$

This integration stage ensures that rapid high-frequency optical noise is filtered out, establishing a stable deterministic precondition $B \in \{0, 1\}$.

## 3 Locality Construction and Causal Boundaries

To prevent superluminal signaling across the optical bench and temporal mirror interfaces, locality construction must be rigorously enforced within the circuit topology. We define the spacetime manifold as $\mathcal{M} = \mathbb{R}^{3,1}$ with metric tensor $g_{\mu\nu}$.

### 3.1 Light-Cone Preservation and Retarded Potentials

Let spatial coordinates across the InGaAs detector array be denoted by $\mathbf{x} \in \mathbb{R}^2$ and temporal coordinates by $t$. The propagation of optical wavefronts from the infrared diode laser through the spatial light modulator to Detectors A and B satisfies the wave equation:

$$\left(\nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right) \mathbf{A}(\mathbf{x}, t) = -\mu_0 \mathbf{J}_{\text{eff}}(\mathbf{x}, t) \tag{7}$$

where $\mathbf{J}_{\text{eff}}$ represents the effective induced current density within the optoelectronic conversion layer.

**Theorem 3.1** (Causal Separation Bound)**.** *For any two computational nodes $i$ and $j$ exchanging state information via the temporal mirror bus with separation vector $\Delta\mathbf{x}_{ij}$ and temporal delay $\Delta t_{ij}$, physical causality is preserved if and only if the interval is time-like or null:*

$$ds^2 = -c^2(\Delta t_{ij})^2 + |\Delta\mathbf{x}_{ij}|^2 \leq 0 \tag{8}$$

*Proof.* The FPGA DDR4 FIFO buffer and JESD204B high-speed serial transceivers operate at clock frequencies bounded by electromagnetic propagation speeds. The propagation delay $\Delta t_{ij}$ across the 16-lane serial bus operating at 12.5 Gbps satisfies $|\Delta\mathbf{x}_{ij}|/\Delta t_{ij} = v_p < c$, ensuring that no closed causal loops violate relativistic microcausality in the laboratory frame. □

# 4 The Inverted Dual Space Manifold

The temporal mirror transformation $\mathcal{M}$ coupled with the optoelectronic viewport inversion operator $\mathcal{V}$ defines a mapping between standard phase space and an *inverted dual space manifold* $\mathcal{Z}^*$.

## 4.1 Symplectic Manifold Inversion

Let the forward computational phase space be coordinatized by canonical variables $(\mathbf{q}, \mathbf{p}) \in \mathcal{Z}$, representing state intensities and momentum-equivalent optical gradients. The mirror transformation $\mathcal{M}$ implements a time-reversal involution combined with spatial coordinate inversion:

$$\mathcal{M} : (\mathbf{q}, \mathbf{p}, t) \longmapsto (\mathbf{q}^*, -\mathbf{p}^*, -t) \tag{9}$$

The inverted dual space manifold $\mathcal{Z}^*$ is constructed via the symplectic form $\omega = d\mathbf{q}^* \wedge d\mathbf{p}^*$.

**Definition 4.1** (Inverted Dual Manifold)**.** *A dual space manifold $\mathcal{Z}^*$ is said to be an inverted dual manifold under $\mathcal{V}$ if the pushforward of the oracle vector field $\mathbf{X}_\mathcal{O}$ satisfies:*

$$\mathcal{V}_* (\mathbf{X}_\mathcal{O}) = -\mathbf{X}_F \tag{10}$$

*where $\mathbf{X}_F$ is the vector field governing forward computation $F$.*

This dual-space construction guarantees that the volume element in phase space undergoes exact determinant inversion:

$$\int_{\mathcal{Z}^*} d\mathbf{q}^* d\mathbf{p}^* = -\int_{\mathcal{Z}} d\mathbf{q} d\mathbf{p} \tag{11}$$

reflecting the convergence of terminal entropy states back into low-entropy initial conditions via optical phase conjugation.

# 5 Time-Reversal ($T$) Symmetry Violation in Recursive Feedback Loops

Fundamental microscopic physical laws (e.g., Maxwell's equations and the Schrödinger equation) are invariant under Time-Reversal symmetry ($T$). However, macroscopic dissipative elements, thresholding components (such as the NPN transistor base-emitter junction and the 74HC74 clocked D flip-flop), and information-erasure feedback loops break $T$-symmetry.

### 5.1 Dissipative Dynamics and Entropy Generation

Consider the time-reversal operator $\mathcal{T}$ acting on state vector $\Psi(t)$. Under microscopic time reversal, momentum and magnetic fields invert ($\mathcal{T}\mathbf{p} = -\mathbf{p}$), whereas spatial coordinates remain invariant ($\mathcal{T}\mathbf{q} = \mathbf{q}$).

In our circuit, the capacitor discharge phase and thresholding non-linearities introduce non-conservative terms into the Lagrangian description. Let the effective Lagrangian of the optoelectronic feedback loop be:

$$\mathcal{L}(\mathbf{q}, \dot{\mathbf{q}}, t) = \frac{1}{2} m \dot{\mathbf{q}}^2 - V(\mathbf{q}) - \gamma \mathbf{q} \cdot \dot{\mathbf{q}} \text{sgn}(\dot{\mathbf{q}}) \tag{12}$$

where $\gamma > 0$ represents the damping coefficient associated with the $10\,\mu$F low-pass smoothing node and transistor conduction losses.

**Theorem 5.1** (Explicit $T$-Symmetry Violation)**.** *The operator equation of motion derived from $\mathcal{L}$ under time-reversal transformation $\mathcal{T}$ yields:*

$$\mathcal{T}\left[m\ddot{\mathbf{q}} + \nabla V(\mathbf{q}) + \gamma \dot{\mathbf{q}} sgn(\dot{\mathbf{q}})\right] = m\ddot{\mathbf{q}} + \nabla V(\mathbf{q}) - \gamma \dot{\mathbf{q}} sgn(\dot{\mathbf{q}}) \neq 0 \tag{13}$$

*demonstrating explicit breaking of $T$-symmetry due to velocity-dependent dissipative coupling.*

*Proof.* Under $\mathcal{T}$, time $t \to -t$, implying velocity $\dot{\mathbf{q}} \to -\dot{\mathbf{q}}$ and acceleration $\ddot{\mathbf{q}} \to \ddot{\mathbf{q}}$. Substituting these into the Euler-Lagrange equations yields a sign-flip on the frictional dissipative term $\gamma \dot{\mathbf{q}} \text{sgn}(\dot{\mathbf{q}})$. Because the physical system must obey forward-time dissipation according to the second law of thermodynamics ($\Delta S_{\text{univ}} > 0$), the reversed equation fails to describe valid physical trajectories unless external work is injected. Thus, the recursive feedback loop containing threshold logic and clocked latches breaks $T$-symmetry dynamically. □

## 6 Conclusion

The Reversed-Process Complexity Circuit establishes an analytical framework merging optoelectronic driver dynamics, relativistic locality bounds, inverted dual-space symplectic manifolds, and rigorous $T$-symmetry violation proofs. By replacing standard feed-forward execution with temporal mirror feedback and oracle-driven initialization, the architecture opens new avenues for exploring non-conservative computational complexity and recursive state convergence.

## References


[1] Bennett, C. H. (1973). Logical Reversibility of Computation. *IBM Journal of Research and Development*, 17(6), 525–532.

[2] Landauer, R. (1961). Irreversibility and Heat Generation in the Computing Process. *IBM Journal of Research and Development*, 5(3), 183–191.